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SYSTEMS AND METHODS FOR RETRACTABLE WHEELS

Abstract

Retractable wheels within a suitcase that are configured to be positioned within the suitcase responsive to applying a downward force on the suitcase.

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Background/Summary

BACKGROUND INFORMATION

Field of the Disclosure

[0001] Examples of the present disclosure are related to systems and methods for retractable wheels. More specifically, embodiments relate to retractable wheels within a suitcase with a gate and a lock. Responsive to moving the lock, the gate may rotate downward to allow a castor to automatically move downward.

Background

[0002] The use of towable baggage has been prevalent, such as suitcases, backpacks, etc. with wheels. Conventionally, towable baggage includes permanently exposed wheels on the bottom of the baggage. However, situations may arise when the towable baggage needs to be stored, and the exposed wheels may occupy too much space. This led to towable baggage systems with wheels that have to be manually retracted or extended. Yet, this can become an arduous and time consuming task, especially when the towable baggage is storing fragile or heavy items.

[0003] Other towable baggage systems include complex systems of locks and springs to retract the wheels. These complex systems require substantial space within the towable baggage, which limits the amount of items that can be stored within the towable baggage.

[0004] Accordingly, needs exist for more effective and efficient systems and methods for a retractable wheel system configured to retract and extend wheels within towable baggage.

SUMMARY

[0005] Embodiments disclosed herein describe systems and methods for a retractable wheel system within towable baggage. The retractable wheel system may be configured to allow the wheels of the baggage to be retractable within the baggage by a user applying downward force against the baggage to compress a spring, and a locking pin on a shaft of the wheel traversing solenoid to position the wheel and the shaft within a housing. To position the wheel outside of the housing, the solenoid may be unlocked and the spring may automatically elongate to apply a force against an upper surface of the shaft to push the wheel outside of the housing. This system may minimize an amount of space occupied by the wheel while stored within the towable baggage, while selectively locking the castor in place within the towable baggage.

[0006] The retractable wheel system may include a wheel, a shaft, spring, housing, and solenoid.

[0007] The wheel may be a castor that is configured to enable an object, such as towable baggage, to be moved. The wheel may be formed of various materials, shapes, and sizes. In a first mode of operation, the wheel may be configured to be positioned outside of the housing, and in a second mode of operation the wheel may be configured to be entirely positioned in the housing. The wheel may include a threaded tube, which is configured to be coupled with an inner circumference of the shaft. In embodiments, responsive to the shaft moving along a linear axis, the wheel may move along the linear axis. This may enable the wheel to be positioned outside of the housing in the first mode, and inside of the housing in the second mode.

[0008] The shaft may be tube, cylinder, projection, etc. having a first end coupled to the wheel, and a second end coupled to the spring. The shaft may be configured to translate forces applied by the spring to the wheel, which may enable the wheel to move from the first mode to a second mode. The shaft may include a locking pin. The locking pin may be a projection, outcrop, protrusion, etc. that is fixed on an outer circumference of the shaft, wherein the locking pin may be configured to extend in a plane orthogonal to a central axis of the shaft. In embodiments, responsive to the spring compressing a vertical offset between the locking pin and a proximal end of the housing may decrease. This may enable the locking pin to traverse the solenoid, and be positioned between the proximal end of the housing and the solenoid. Responsive to the spring elongating, the locking pin may traverse the solenoid in an opposite direction, and be positioned between a distal end of the housing and the solenoid. In embodiments, the locking pin may be positioned along the shaft such that the locking pin is always positioned within the housing, which may limit snagging or breaking

of the locking pin.

[0009] The spring may be a device that is configured to compress and elongate, while applying a constant spring force against the shaft. The spring may have a first end that is coupled with the proximal end of the housing, and a second end that is coupled with an upper surface of the shaft. The constant spring force may be in a first direction applied from the proximal end of the housing towards the distal end of the housing, and be applied against the shaft. In embodiments, the constant spring force applied by the spring against the shaft may maintain the wheel in the second mode until an external force is applied against the towable baggage in a second direction that is greater than the spring force. For example, a user may apply a downward force against the towable baggage to create the external force. Responsive to the external force compressing the spring, the locking pin may traverse the solenoid, and lock the wheel within the housing. When the solenoid is locking the locking pin in place, the spring may apply the constant spring force in the first direction. However, the locked solenoid may not enable the spring to elongate. Responsive to unlocking the solenoid, the constant spring force may move the locking pin to be positioned between the solenoid and the distal end of the housing, which may allow the wheel to be positioned outside of the housing. This may enable the wheels to move while the spring is elongated and applying the constant spring force against the shaft, which may give the wheels some shock protection.

[0010] The housing may be a device that is configured to hold and secure the elements of the retractable wheel system. The housing may be configured to be embedded within the towable storage system or luggage. The housing may have a distal end that is positioned flush with a lower surface or side surface of the towable storage. A proximal end of the housing may be positioned within the towable storage, and a first end of the spring may be positioned against the proximal end. Between the proximal end and the distal end of the housing may be a chamber, wherein the spring may elongate and compress within the chamber. Based on the elongation and compression of the spring, the shaft and wheel may also move axially within the chamber. The housing may also include a solenoid.

[0011] The solenoid may be a device that is configured to selectively lock the locking pin in place. The solenoid may be configured to modify a size of the inner diameter of the chamber to be greater than or smaller than an outer diameter of the locking pin. Furthermore, the solenoid may be a one-way gate that is configured to rotate upward or otherwise increase the inner diameter across the solenoid to allow the locking pin to always move upward. After the locking pin passes the solenoid, the solenoid may automatically decrease the inner diameter across the chamber, which may restrict the downward and upward movement of the locking pin. Responsive to the solenoid becoming unlocked, the solenoid may rotate downward or otherwise increase the inner diameter across the solenoid to allow the locking pin the selectively move downward or upward. After the locking pin passes the solenoid, the solenoid may automatically decrease the inner diameter of the chamber.

[0012] Accordingly, the solenoid may be a locking mechanism configured to allow for the automatic upward movement of the locking pin or the upward movement of the locking pin while the solenoid is activated, past the solenoid, while selectively locking the locking pin above the solenoid. In embodiments, responsive to receiving an input from the user, the solenoid may be retracted or otherwise increase an inner diameter across the chamber, which may allow the locking pin to move downward past the solenoid. To this end, the solenoid may be a one-way gate that is configured freely allow the upward movement of the locking pin or allow the upward movement of the locking pin once the solenoid is unlocked, but may be configured to be selectively locked in a plane extending orthogonal to a central axis of the shaft. The solenoid may create a profile on an inner diameter of the housing, which may selectively decrease the inner diameter of the housing. Responsive to the locking pin moving from a distal end of the chamber towards the proximal end of the chamber, the locking pin may apply a force against a lower surface of the solenoid, moving the solenoid upward, and increasing a size across the inner diameter of the chamber. Due to the

increase in size across the inner diameter of the chamber, the locking pin may move to a position between the solenoid and the proximal end of the housing. When the locking pin moves past the solenoid, the solenoid may return to the plane extending orthogonal to a central axis of the housing. The locking pin may then rest on an upper surface of the solenoid.

[0013] In embodiments, because the solenoid is selectively locked in place, the solenoid may not move downward until desired. To this end, an actuator or mechanical device may apply a force (or remove a force) from the solenoid, allowing the solenoid to swing downward. This may allow the spring to apply a downward force, causing the locking pin to translate this force to the solenoid, allowing the solenoid to swing downward, and move the wheel outside of the housing. Responsive to the external force being applied to the wheels, the locking pin may move upward, causing the solenoid to swing upward, and automatically return to its locked position.

[0014] These, and other, aspects of the invention will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. The following description, while indicating various embodiments of the invention and numerous specific details thereof, is given by way of illustration and not of limitation. Many substitutions, modifications, additions or rearrangements may be made within the scope of the invention, and the invention includes all such substitutions, modifications, additions or rearrangements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.

[0016] FIG. 1 depicts towable baggage, according to an embodiment.

[0017] FIG. 2 depicts the upper surface of towable baggage, according to an embodiment.

[0018] FIG. 3 depicts the front surface of towable baggage, according to an embodiment.

[0019] FIG. 4 depicts a retractable wheel system, according to an embodiment.

[0020] FIG. 5 depicts a retractable wheel system, according to an embodiment.

[0021] FIG. 6 depicts a side view of a retractable wheel system, according to an embodiment.

[0022] FIG. 7 depicts a side view of manual unlock assembly, according to an embodiment.

[0023] FIG. 8 illustrates a method for retractable wheels in towable baggage.

[0024] FIG. 9 depicts an embodiment of a solenoid or locking system in the locked position, wherein locking system is configured to allow the selective movement of the locking pin.

[0025] FIG. 10 depicts an embodiment of a solenoid or locking system in the unlocked position, wherein locking system is configured to allow the selective movement of the locking pin.

[0026] Corresponding reference characters indicate corresponding components throughout the several views of the drawings. Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present disclosure. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0027] In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present embodiments. It will be apparent, however, to one having ordinary skill in the art that the specific detail need not be employed to practice the present embodiments. In other instances, well-known materials or methods have not been described in

detail in order to avoid obscuring the present embodiments.

[0028] Embodiments disclosed herein describe systems and methods for a retractable wheel system within towable baggage. The towable baggage may be any type of container, such as luggage, a backpack, etc. that is configured to be moved between locations. The towable baggage may include a plurality of retractable wheel systems, which may be positioned at conventional locations associated with towable baggage. For example, the towable baggage may have retractable wheel systems positioned on a lower surface of the towable baggage.

[0029] FIG. 1 depicts towable baggage **100**, according to an embodiment. Towable baggage **100** may include a retractable handle **110**, collapsible cup holder **120**, plurality of locks **130**, **132**, **132**, and recessed handle **110**.

[0030] Retractable handle **110** may be a handle positioned on an upper surface of towable baggage **100** that is configured to elongate to be positioned outside of towable baggage **100**, and retracted to be positioned within towable baggage **100**. In embodiments, retractable handle **110** may include a button lock, which may be configured to selectively lock the retractable handle in the retracted position. Responsive to a user pressing the button and applying forces against the retractable handle **110** in an upward direction, the retractable handle **110** may move from the retracted position to the elongated position.

[0031] Collapsible cup holder **120** may be a cup holder with a cylindrical top opening that is configured to receive a cup, drink, cylindrical vessel when in a use mode, wherein in the use mode the cylindrical top opening may extend in a direction orthogonal to a central axis of the towable baggage **100**. Responsive to a user applying a downward force against the top of the collapsible cup holder **120**, the cylindrical top opening may rotate downward, such that the cylindrical top opening is positioned in parallel to the central axis of the towable baggage.

[0032] Towable baggage **100** may include a plurality of locks **130**, **132**, **134** that are configured to secure compartments within towable baggage **100**. Locks **130**, **132**, **134** may be TSA-accepted locks that can be opened by authorities using publically known universal master keys. In embodiments, locks **130**, **132**, **134** may include an indicator that will appear if opened by the master key. This may enable the owner of towable luggage **100** to know which lock, if any, was opened.

[0033] Recessed handle **140** may be a handle positioned on the upper surface of towable baggage **100**. Handle **140** may be made of cloth and be configured to extend away from the upper surface. This may be convenient when it is not practical to extend retractable handle **110**, such as when loading towable baggage into a vehicle.

[0034] FIG. 2 depicts the upper surface of towable baggage **100**, according to an embodiment. Elements depicted in FIG. 2 may be described above, and for the sake of brevity another description of these elements may be omitted.

[0035] As depicted in FIG. 2, when collapsible cup holder **120** is in a storage position, the cylindrical opening may be positioned flush against the rear surface of towable baggage **100** in parallel to a central axis of towable baggage **100**. This may reduce the profile of towable baggage.

[0036] Furthermore, as depicted, the upper surface of towable baggage **100** may include a top identification tag **210**, and a front surface of towable baggage **100** may include a front identification tag **220**. Identification tags **210**, **220** may include information associated with the user of towable baggage. In embodiments, the identification tags **210**, **220** may be customizable and removable identification tags.

[0037] Additionally, the front surface of towable baggage **100** may include a front zipper slot **230**. Front zipper slot **230** may be utilized to open and close a front compartment of towable baggage **100**.

[0038] FIG. 3 depicts the front surface of towable baggage **100**, according to an embodiment. Elements depicted in FIG. 3 may be described above, and for the sake of brevity another description of these elements may be omitted.

[0039] Front surface **310** may also include a logo **310**. The logo may be utilized for branding purposes, and also for identification purposes of towable baggage **100**. Positioned between front surface **310** and a rear surface may be hinge **320**. Hinge **320** may be configured to allow front surface **310** to be rotated away from the rear surface to expose a central compartment within towable baggage.

[0040] Front surface **310** may also include a plurality of rivets **330**, which are configured to secure a wheel housing of towable luggage **100**.

[0041] Front surface **310** may also include a manual unlock assembly **340**. The manual unlock assembly may be configured to receive external forces from a user to move an actuator between a locked to unlock position, wherein in the locked position the wheels may be maintained within towable baggage **100**, and when the actuator is in the unlocked position the wheels may automatically extend to a position outside of towable baggage **100**.

[0042] FIG. **4** depicts a retractable wheel system **400**, according to an embodiment. Retractable wheel system **400** may be positioned on a lower surface of towable baggage **100**, wherein towable baggage **100** may have a plurality of retractable wheel systems **400**. For example, a piece of luggage may have four retractable wheels systems **400** that are each independently controlled. Retractable wheel system **400** may be configured to allow wheels to freely rotate outside of retractable wheel systems **400**, and when desirable, allow a user to perform actions to lock the wheels within retractable wheel system **400**. Each retractable wheel system **400** may include a wheel **430**, a shaft **440**, spring **450**, housing **410**, and actuator **420**.

[0043] Wheel **430** may be a castor that is configured to enable an object, such as towable baggage **100** to be moved. Wheel **430** may be formed of various materials, shapes, and sizes. In a first mode (use mode) of operation, Wheel **430** may be configured to be positioned outside of the housing **410**, **412**, and in a second mode (storage mode) of operation, wheel **430** may be configured to be positioned in the housing **410**, **412**. Wheel **430** may include a threaded tube **432**, which is configured to be coupled with an inner circumference of the shaft **440**. In embodiments, responsive to the shaft **440** moving along a linear axis, wheel **430** may move along the linear axis. This may enable wheel **430** to be positioned outside of the housing **410**, **412** in the first mode, and inside of the housing **410**, **412** in the second mode.

[0044] The shaft **440** may be tube, cylinder, projection, etc. having a first end coupled to the wheel **430**, and a second end positioned directed adjacent to spring **450**. Shaft **440** may be configured to translate forces applied by the spring **450** to the wheel **430**, which may enable the wheel **430** to move from the first mode to a second mode. Shaft **440** may include a locking pin **442**. Locking pin **442** may be a projection, outcrop, protrusion, etc. that is fixed on an outer circumference of the shaft **440**, wherein locking pin **442** may be configured to extend in a plane orthogonal to a central axis of the shaft **440**. Locking pin **442** may have a fixed outer diameter that is greater in size than the outer diameter of shaft **440**. In embodiments, locking pin **442** may be configured to selectively secure wheel **430**. Specifically, when locking pin **442** is secured within housing **410**, **412** via a gate, wheel **430** may be positioned within housing **410**, **412**. When the movement of locking pin **442** is no longer restricted by the gate, wheel **430** may be positioned outside of the housing **410**, **412**.

[0045] Spring **450** may be a device that is configured to compress and elongate, while applying a constant spring force, wherein the constant spring force may be in a downward direction towards shaft **440**, which may be orthogonal to the central axis of locking pin **442**. Spring **450** may have a first end that is coupled with the proximal end **414** of housing **410**, **412**, and a second end that is coupled with the distal end **416** of housing **410**, **412**. In embodiments, the constant spring force applied by the spring **450** against the shaft may maintain the wheel **430** in the second mode until an external force is applied against the towable baggage **100** and spring **450** in a second direction that is greater than the spring force. For example, a user may apply a downward force against the upper surface of towable baggage **100** to create the external force. Responsive to the external force compressing the spring **450** while the solenoid is able to be unlocked, the locking pin **442** may

traverse the gate or solenoid (not shown), and the gate or solenoid may lock the wheel **430** within the housing **410**, **412**. In embodiments, responsive to the spring **450** compressing a vertical offset between the locking pin **442** and a proximal end **414** of the housing **410**, **412** may decrease. This may enable the locking pin **442** to traverse the solenoid, and be positioned between the proximal end **414** of the housing **410**, **412** and the solenoid along the central axis of retractable wheel system **400**. When the solenoid is locking the locking pin **442** in place, the spring **450** may apply the constant spring force in the first direction. However, the locked solenoid may not enable the spring **450** to elongate.

[0046] Responsive to the spring **450** elongating, the locking pin **442** may traverse the solenoid in an opposite direction, and be positioned between a distal end **414** of the housing **410**, **420** and the solenoid. In other words, when the solenoid is unlocked, the constant spring force may move the locking pin **442** to be positioned between the solenoid and the distal end **416** of the housing **410**, **412**, which may allow the wheel **430** to be positioned outside of the housing **410**, **412**. When the wheels **430** are positioned outside of housing **410**, **420**, the towable baggage **100** may be rolled while the spring **450** is elongated and applying the constant spring force against the shaft **440**, which may give the wheels **430** some shock protection.

[0047] Housing **410**, **412** may be a device that is configured to hold and secure the elements of the retractable wheel system **400**. The housing may be configured to be embedded within the towable storage system **100** or luggage. Housing **410**, **412** may have a distal end **416** that is positioned flush with a lower surface or side surface of the towable baggage **100**. Housing **410**, **412** may also include a proximal end **414** positioned within the towable baggage **100**, and a first end of the spring **450** may be positioned against the proximal end **414**. Between the proximal end **414** and the distal end **416** of the housing **410**, **412** may be a chamber (shown in FIG. 5), wherein the spring **450** may elongate and compress within the chamber. Based on the elongation and compression of the spring **450**, shaft **440** and wheel **430** may also move axially within the chamber. To control the elongation and compression of spring **450**, housing **410**, **412** may include a solenoid.

[0048] The gate or solenoid may be a device that is configured to selectively lock the locking pin **442** in place. Generally, the gate or solenoid may be any mechanical device that is configured to selectively change an inner diameter across the chamber. When the inner diameter across the chamber is larger than an outer diameter associated with the locking pin **442**, spring **450** may compress or elongate. However, when the inner diameter across the chamber is smaller than the outer diameter associated with locking pin **442**, spring **450** may not transition between a compressed an elongated state.

[0049] In embodiments, the gate or solenoid may be configured to allow for the upward movement of locking pin **442**, past the gate or solenoid, while selectively locking the locking pin **442** above the gate or solenoid. This may allow for spring **450** to freely compress, while selectively restricting the elongation of spring **450**.

[0050] In embodiments, responsive to receiving an input from the user, the gate or solenoid may be retracted or otherwise increase an inner diameter across the chamber. This may allow the locking pin **442** to move downward past the solenoid. To this end, the gate or solenoid may be a one way gate that is configured freely allow the upward movement of the locking pin **442**, wherein forces applied by locking pin **442** against the gate or solenoid may cause the inner diameter across the solenoid to increase. However, the gate or solenoid may not freely rotate when downward forces are applied to the upper surface of the solenoid by the locking pin **442**. Accordingly, the gate or solenoid may be configured to be selectively locked in a plane extending orthogonal to a central axis of the shaft **440**, with the solenoid creating an inner diameter across the chamber to be smaller than the outer diameter associated with locking pin **442**.

[0051] Responsive to the locking pin **442** moving from distal end **416** of the chamber towards the proximal end **414** of the chamber, locking pin **442** may apply a force against a lower surface of the solenoid, moving the solenoid upward, and increasing a size across the inner diameter of the

chamber. Due to the increase in size across the inner diameter of the chamber, locking pin **442** may move to a position between the gate or solenoid and the proximal end **414** of the housing **410**, **412**. When locking pin **442** moves upward past the gate or solenoid, the gate or solenoid may automatically return to the plane extending orthogonal to a central axis of the housing. However, because the gate or solenoid cannot freely rotate downward, locking pin **442** may then rest on an upper surface of the gate or solenoid. In embodiments, because the gate or solenoid is selectively locked in place, the gate or solenoid may not move downward until desired. To this end, an actuator or mechanical device may apply a force (or remove a force) from the gate or solenoid, allowing the gate or solenoid to swing downward or otherwise reduce an inner diameter across the chamber. This may allow the spring **450** to apply a downward force, causing locking pin **442** to translate this force to the gate or solenoid, allowing the gate or solenoid to swing downward or increase an inner diameter across the chamber, and allowing the wheel **430** to move outside of the housing. Responsive to the external force being applied to wheels **430**, locking pin **442** may move upward, causing the solenoid to swing upward or otherwise increase the inner diameter across the chamber, and automatically return to its locked position extending in a direction orthogonal to shaft **440** and having an inner diameter that is smaller than the outer diameter of locking pin **442**.

[0052] In embodiments, the gate or solenoid may be selectively unlocked, allowing spring **450** to transition from a compressed state to an elongated state based on a user interacting with towable baggage **100**. The user's interaction may be to press a button, such as button **422**, to mechanically release the gate or solenoid, allowing the gate or solenoid to rotate downward or retract. Alternatively, button **422** may be utilized to remove an electric charge associated with the gate or solenoid, which may allow the gate or solenoid to rotate downward or retract, and once button **422** is no longer pressed, then the solenoid may automatically rotate or move back to its locked position.

[0053] FIG. **5** depicts a retractable wheel system **400**, according to an embodiment. Elements depicted in FIG. **5** may be described above, and for the sake of brevity another description of these elements may be omitted.

[0054] As depicted in FIG. **5**, spring **450** may be positioned within a chamber **520** of the housing. Spring **450** may have a first end permanently positioned against proximal end **414** at a fixed location. A second end of spring **450** may be configured to be positioned within a recess **510** on a proximal end of shaft **440**. Responsive to spring **450** elongating or compressing, a distance between the first and second ends of spring **450** may change.

[0055] FIG. **6** depicts a side view of retractable wheel system **400**, according to an embodiment. Elements depicted in FIG. **6** may be described above, and for the sake of brevity another description of these elements may be omitted.

[0056] FIG. **7** depicts a side view of manual unlock assembly **340**, according to an embodiment. Elements depicted in FIG. **7** may be described above, and for the sake of brevity another description of these elements may be omitted.

[0057] In embodiments, each retractable wheel assembly may include a manual unlock assembly **340**, wherein a corresponding manual unlock assembly **340** may be configured to manually unlock a solenoid. This may allow the locking pin **420** to no longer be positioned on the gate or solenoid and allow the spring **450** to elongate. Unlock assembly **340** may include a base **710**, spring **715**, and button **720**.

[0058] The base **710** may be configured to be positioned flush on a front or rear surface of the towable baggage. Spring **715** may be configured to elongate and compress to allow a front face of button **720** to be naturally positioned substantially flush with a front face of base **710**. Responsive to a user pressuring button **720** inward, spring **715** may compress allowing an inner surface of button **720** to interact with the gate or solenoid to unlock the gate or solenoid. Responsive to the user no longer pressing the button, spring **715** may push button outward.

[0059] To this end, the gate or solenoid may be manually or electronically locked in place.

[0060] FIG. 8 illustrates a method **800** for retractable wheels in towable baggage. The operations of method **800** presented below are intended to be illustrative. In some embodiments, method **800** may be accomplished with one or more additional operations not described, and/or without one or more of the operations discussed. Additionally, the order in which the operations of method **800** are illustrated in FIG. 8 and described below is not intended to be limiting.

[0061] At operation **810**, wheels may be extended from a housing, and allow baggage to be rolled. When the wheels are extended, a spring within a housing may be able to slightly compress and elongate within locking the wheels within the housing.

[0062] At operation **820**, a user may perform actions the towable baggage to compress the spring. For example, a user may push down on an upper surface of the towable baggage.

[0063] At operation **830**, due to the spring compressing a locking pin may traverse a gate or solenoid, wherein the gate or solenoid may have a smaller inner diameter than that of the locking pin when the locking pin is crossing the gate or solenoid. In embodiments, the locking pin may physically touch and move the solenoid to change an inner diameter across the solenoid. For example, the gate or solenoid may be a one way hinge that can freely rotate in an upward angle.

[0064] At operation **830**, once the locking pin passes the gate or solenoid, the gate or solenoid may automatically revert back to a locked position, and extended in a direction orthogonal to a central axis of the spring. When in the locked position, the inner diameter across the gate or solenoid may be smaller than that of the locking pin.

[0065] At operation **850**, the user may no longer apply force against the upper surface of the towable baggage, and the spring may partially elongate. The spring may elongate a distance until a lower surface of the locking pin is positioned against the locked solenoid, wherein the locked solenoid may secure the wheel within the housing.

[0066] At operation **860**, the user may perform actions to unlock the gate or solenoid. For example, the user may press a button to mechanically unlock the solenoid, or press a button to electronically unlock the gate or solenoid. When the gate or solenoid is unlocked, a force created by the locking pin against the upper surface of the gate or solenoid may cause the solenoid to have a larger inner diameter than that of the outer diameter of the locking pin. This may allow the locking pin to be moved downward due to the force of the spring, and the wheels to be positioned outside of the housing.

[0067] As such, systems described herein allow for wheels to be retracted within a housing responsive to a user performing actions on a user interface and simultaneously providing a compressive force against a spring. Responsive to the user performing further actions on the user interface, the wheels may automatically become extended.

[0068] FIG. 9 depicts an embodiment of a solenoid or locking system **900** in the locked position, wherein locking system **900** is configured to allow the selective movement of the locking pin. Locking system **900** may include gate **910**, lock **920**, button **422**, and a solenoid.

[0069] Gate **910** may be configured to rotate upwards responsive to receiving a force, and be selectively locked in place on lock **920**. Gate **910** may also be configured to selectively rotate downward responsive to moving lock **920**. Gate **910** may include a first arm **912**, second arm **914**, and channel **916**. First arm **912** may have a first length, which is shorter than a second length of second arm **912**. First arm **912** may be configured to extend across a slot **905** within the housing when in the locked position, and not extend across the slot **905** when in the unlocked position. When in the locked position, a locking pin **442** may be configured to be positioned on an upper surface of first arm **912**, within channel **916**, and within slot **905**.

[0070] Second arm **916** may have a length that is always configured to extend across slot **905**, which may limit the upward movement of locking pin **442**. When in the locked position, a profile **918** of the distal end of arm **916** may rest on a ledge **922** on lock **920**. The interfacing of profile **918** and ledge **922** may limit the downward movement of second arm **916**, which may maintain gate **910** in the locked position until an external force is applied against button **422**. Furthermore, when

in the locked position, the upward movement of second arm **916** may be restricted due to a profile on the inner diameter of lock **924**.

[0071] Lock **920** may be configured to selectively secure second arm **920** extending across slot **905**, which may maintain locking pin **442** and the wheels within the housing. When in the locked position, lock **920** may extend in a direction perpendicular to an axis of second arm **916**. Lock **920** may be configured to rotate to slide ledge **922** away from profile **918**. Specifically, shaft **931** may pull button **422** to move ledge **922**. This may allow second arm **914** to rotate downward. When second arm **914** rotates downward, first arm **912** may no longer extend across slot **905**, allowing a spring within a housing to apply a downward force against locking pin **442**. The downward force created by the spring may be applied to first arm **912**, forcing the downward rotation of second arm **914**. Subsequently, shaft **931** may push button **422** to move first arm **912** across slot **905** to lock the locking pin **422** in place.

[0072] Button **422** may be positioned on an upper surface of lock **420**. Button **422** may be a shaft that is configured to receive forces from the solenoid to release lock **420**, allowing ledge **922** to move away from profile **918**. Specifically, button **422** may be configured to receive a force from the solenoid to maintain in parallel to a central axis of slot **905**. This may allow ledge **922** to be positioned under profile **918**, and maintain gate **910** in the locked position. When the solenoid applies a force, or removes a force, button **422** may move ledge **922** away from slot **905**, allowing gate **910** to automatically swing downward. Responsive to the solenoid reapplying the force directly against button **422**, button **422** may automatically move ledge **922** towards the central axis of slot **905**, such that button **422** is normally positioned closer to the central axis of slot **905** and only moves away from slot responsive to the solenoid performing actions to pull button **422** inward towards a plane aligned with the central axis of slot **905**, within the housing. In other embodiments, button **422** may directly receive forces from a user to move lock **920** from a position where ledge **922** is under profile **918** to a position where ledge **922** is pulled away from profile **918**.

[0073] The solenoid may be an electronic device that is configured to apply forces or remove forces against button **422**. In embodiments, responsive to the solenoid receiving an electric command, the solenoid may be configured to perform actions such that a shaft **931** directly coupled to button **422** pulls button **422** inward. Accordingly, shaft **931** may pull button **422** towards a central axis of the housing, which may allow gate **910** to move from a locked position to an unlocked position. Responsive to the solenoid no longer performing actions to pull button **422**, a spring may apply a force directly against button **422** to rotate ledge **922** towards the central axis of the housing. This may allow the spring to maintain ledge **922** under profile **918** after the locking pin moves gate **910** to the locked position, wherein ledge **922** may remain under profile **918** until the solenoid creates a force to move shaft **931** and button **422** towards the central axis of housing. This inward movement of button **422** may move ledge **922** outward, allowing gate **910** to become unlocked.

[0074] FIG. **10** depicts an embodiment of a solenoid or locking system **900** in the unlocked position, wherein locking system **900** is configured to allow the selective movement of the locking pin.

[0075] As depicted in FIG. **10**, when ledge **922** is no longer under profile **918**, gate **910** may automatically swing downward, and profile **918** may be positioned adjacent to a lower surface of housing **1010**. When gate **910** swings downward, second arm **914** may be positioned at a downward angle across slot **905** and first arm **912** may be positioned at a downward angle but end before slot **905**. This may enable a lower surface of second arm **914** to receive an upward force from a locking pin **442** to move gate **910** from an unlocked position to a locked position.

[0076] Specifically, when the locking pin **442** applies an upward force against gate **910**, a tapered outer profile **1030** of gate **910** may interface with a tapered inner profile of lock **1020**. This may once again lock the locking pin **442** within slot **905** by outwardly moving ledge **922** until profile **918** is positioned above ledge **922**. Due to the reduction in length across ledge **922**, and the forces

against button **422** to inwardly move ledge **922**, responsive to positioning profile **918** above ledge **922**, ledge **922** may automatically move inward, which may lock gate **910** in place.

[0077] Although the present technology has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred implementations, it is to be understood that such detail is solely for that purpose and that the technology is not limited to the disclosed implementations, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present technology contemplates that, to the extent possible, one or more features of any implementation can be combined with one or more features of any other implementation.

[0078] Reference throughout this specification to “one embodiment”, “an embodiment”, “one example” or “an example” means that a particular feature, structure or characteristic described in connection with the embodiment or example is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment”, “in an embodiment”, “one example” or “an example” in various places throughout this specification are not necessarily all referring to the same embodiment or example. Furthermore, the particular features, structures or characteristics may be combined in any suitable combinations and/or sub-combinations in one or more embodiments or examples. In addition, it is appreciated that the figures provided herewith are for explanation purposes to persons ordinarily skilled in the art and that the drawings are not necessarily drawn to scale.

Claims

1. A wheel assembly for luggage comprising: a locking pin configured to move axially; a gate with a first arm and a second arm configured to control the positioning of the locking pin, the first arm having a first length, and the second arm having a second length, the first length being shorter than the second length, wherein the locking pin is configured to be positioned between the first arm and the second arm when the wheel assembly is locked and the locking pin positioned below the first arm when the wheel assembly is unlocked; a securing mechanism configured to secure the gate in place.
2. The system of claim 1, wherein when the wheel assembly is locked the first arm blocks movement of the locking pin.
3. The system of claim 1, wherein the securing mechanism includes a ledge, the ledge being configured to be positioned under a distal end of the second arm when the wheel assembly is locked, and the ledge being positioned away from the distal end of the second arm when the wheel assembly is unlocked.
4. The system of claim 3, wherein the ledge moves away from the distal end of the second arm to transition the wheel assembly from being locked to unlocked.
5. The system of claim 1, wherein the second arm extends at a downward angle to allow the locking pin to move upward.
6. The system of claim 1, wherein the wheel assembly includes a housing and a spring, the spring being configured to apply a constant downward spring force, wherein the constant downward spring force automatically moves the locking pin downward when the first arm is not extending across the slot.
7. The system of claim 6, wherein when an external force opposite the constant downward spring force the wheel assembly moves from the unlocked position to the locked position.
8. The system of claim 7, wherein when the external force is applied to the locking pin, the locking pin moves vertically along the linear axis to move the second arm of the gate upward.
9. The system of claim 1, further comprising: a suitcase housing the wheel assembly, wherein the suitcase includes four independently operating wheel assemblies.

10. The system of claim 9, further comprising: a cupholder positioned on a rear surface of the suitcase.
